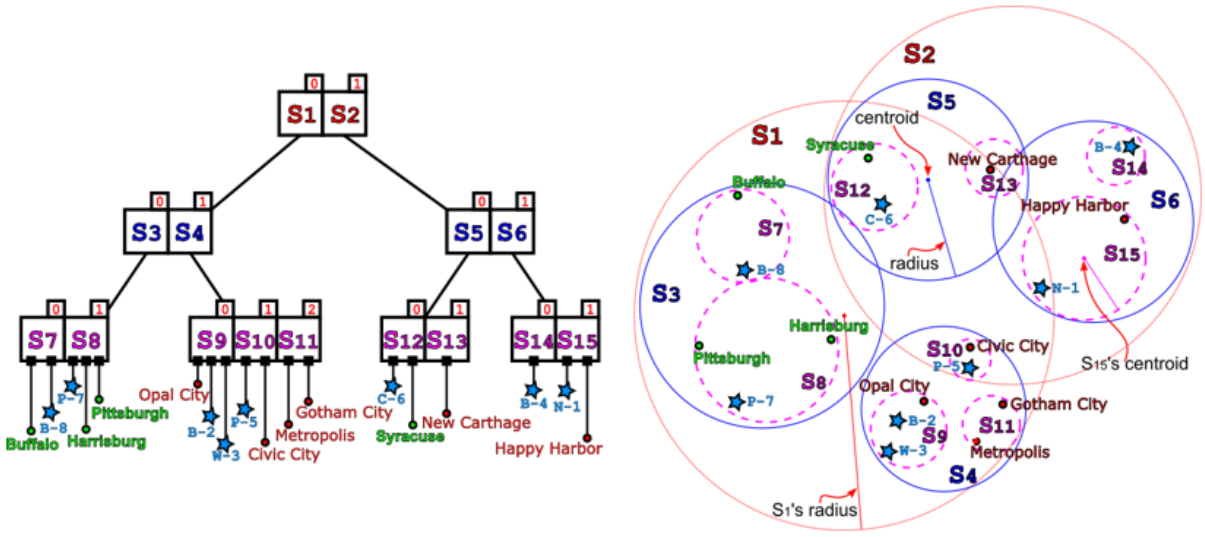


Figure 10.9 Representation of a possible SS-tree covering the same dataset of figures 10.4 and 10.5, with parameters `m==1` and `M==3`. As you can see, the tree structure is similar to R-trees'. For the sake of avoiding clutter, only a few spheres' centroids and radii are shown. For the tree, we use the compact representation (as discussed in figures 10.5 and 10.8).

As for R-trees, SS-trees have two parameters, `m` and `M`, respectively the minimum and maximum number of children each node (except the root) is allowed to have.

And like R-trees, bounding envelopes in a SS-tree might overlap: to reduce the overlap, some variants like *SS+-trees* introduce a fifth property (also used in R-tree's variants like *R*-trees*), another heuristic used on insert that perform major changes to re-structures the tree; we will talk about SS+-trees later in this chapter, but for now we will focus on the implementation of plain SS-trees.